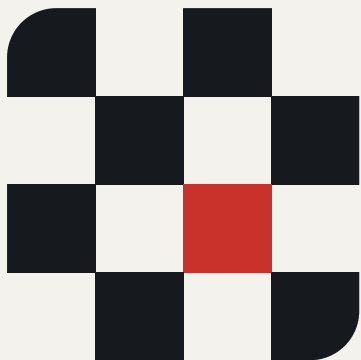


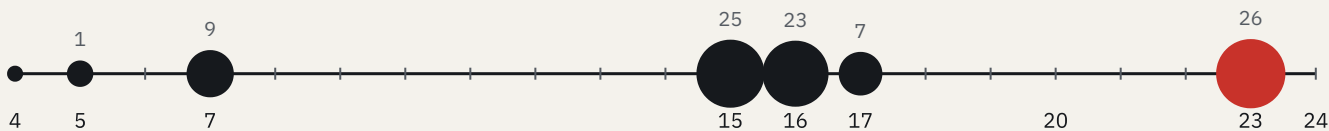


Exchequer

From idea to execution — the project timeline

How a one-page idea for tracing crypto fraud became a tool that follows a victim's money across five blockchains, screens it against six governments' lists, estimates how much of it reached the exchange, and drafts the letter an officer sends.

COMMITTS PER DAY · 4 → 23 SEPTEMBER 2026



20

days, idea to today

91

commits

45,003

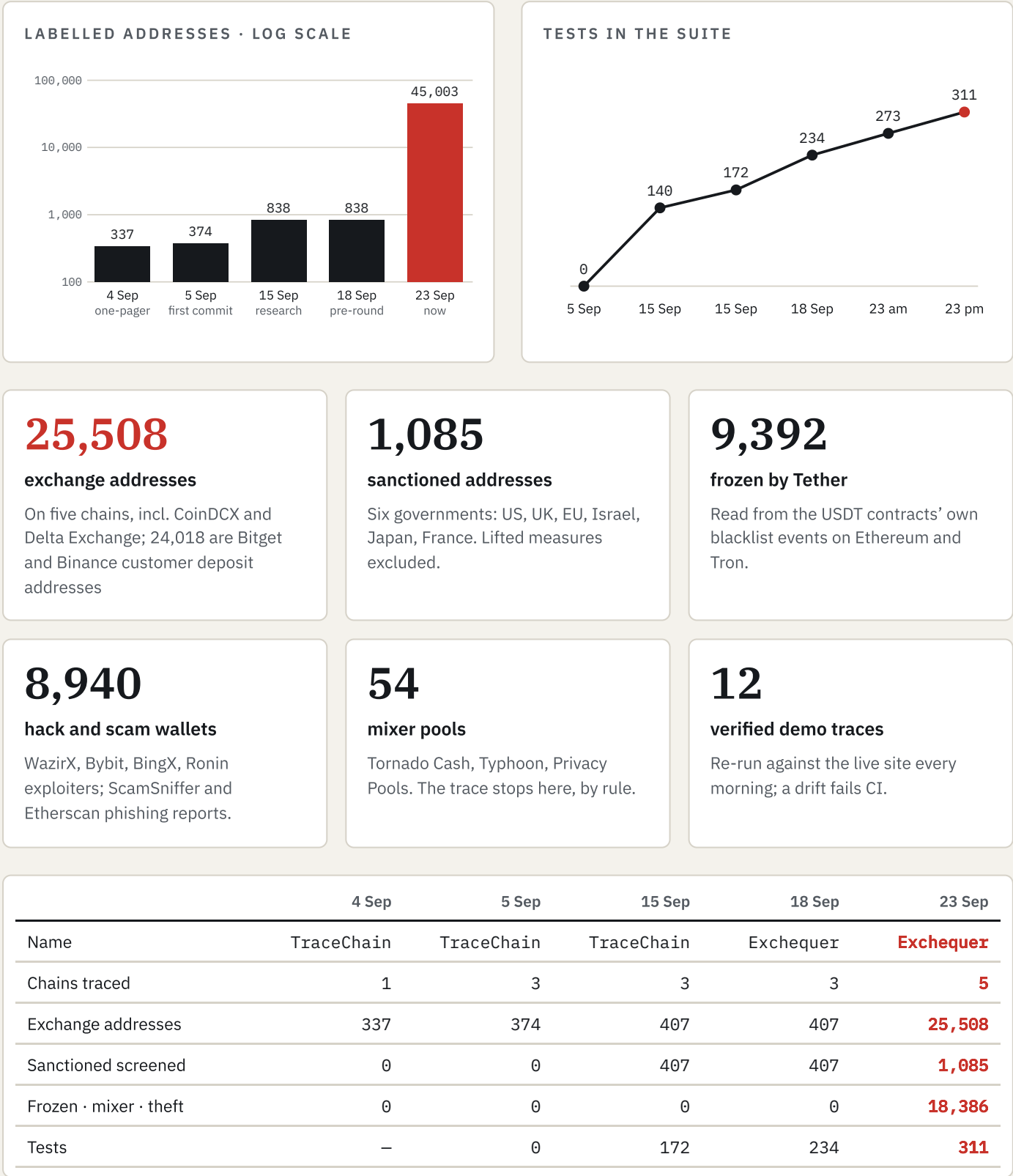
labelled addresses

311

tests, no network

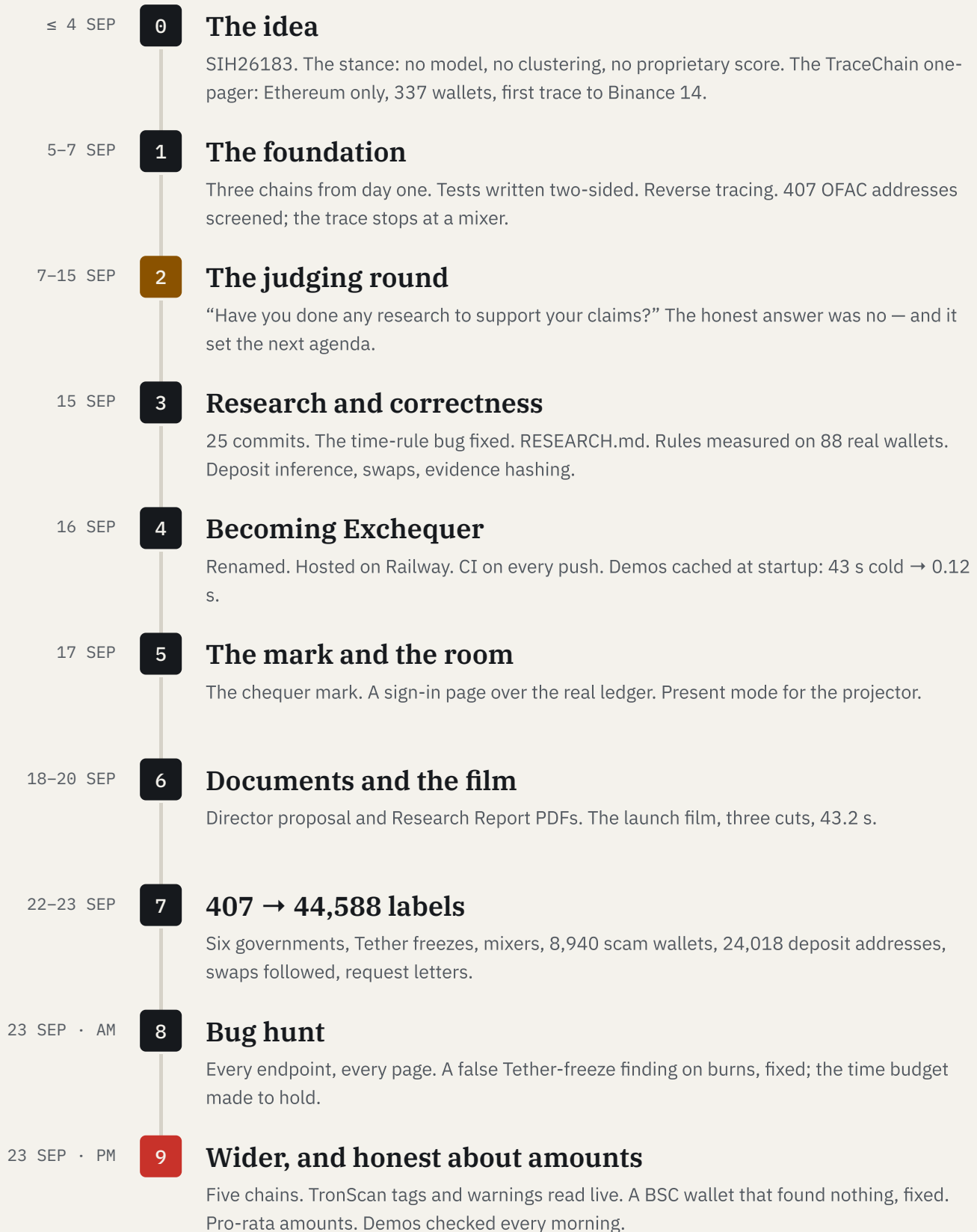
AT A GLANCE

The project in numbers



OVERVIEW

The arc, in ten phases

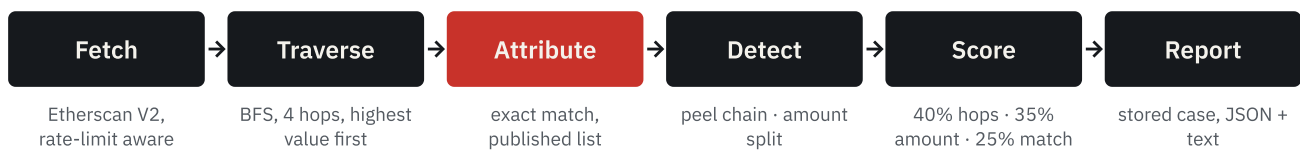


UP TO 15 SEPTEMBER

Phases 0–2 · from an idea to a question

PHASE 0 · THE IDEA · UP TO 4 SEP

SIH26183 asks what a police officer should do with a victim's wallet address. The answer the team chose was a stance before it was code: **no model, no clustering, no proprietary score** — a conclusion that reaches a courtroom must be one an officer can re-check by hand. The one-pager of 4 September laid it out as six steps.



First live result: a reported address traced to **Binance 14 in 2 hops, confidence 0.80**, 138 addresses mapped. 337 verified wallets, 18 exchanges, Ethereum only.

PHASE 1 · THE FOUNDATION · 5–7 SEP

The first commit traced three chains from day one, because Etherscan's free tier serves Ethereum mainnet only. One asset per trace — amounts in different coins cannot be compared or scored.



By 7 September: a two-sided test suite (every rule must fire on its shape *and* stay silent on ordinary activity), reverse tracing to find the other people who paid a scammer, and **407 addresses from OFAC's live sanctions list**. The mixer list came back empty — Tornado Cash had been delisted — and that was written down as the correct answer.

PHASE 2 · THE INTERNAL JUDGING ROUND · BETWEEN 7 AND 15 SEP

“Since you pointed out money laundering, have you done any research to support your claims?”

The honest answer was no. The README asserted that laundering moves as stablecoins, that USDT on Tron is the dominant rail for Indian scam proceeds, and that its thresholds were right — and cited nothing. The same judges would sit the next round. The exact date of this round is not recorded in the repository.

15 SEPTEMBER · 25 COMMITS

Phase 3 · research, correctness and measurement

One long session, run from a written brief with a fixed budget and ordered phases — adversarial review first, evidence second, features after. Tests went from 140 to 172.

The time rule

The one real correctness bug: the trace followed transfers made before the victim's money arrived. Now only later transfers are followed.

RESEARCH.md

Every laundering claim with the source read, the exact passage and a verdict. Two overstated claims rewritten.

Deposit inference

A wallet whose every outflow sweeps to one exchange is named its probable deposit address — the one a request must name.

Swaps at routers

20 Ethereum and 4 BSC routers. A swap's receipt is read so the trace says what the money became.

Tron 7 → 40

Tron labels read from TronScan's own tags and checked on chain: 7 wallets / 4 exchanges → 40 / 18.

Related cases

Stored cases whose money converges. Three phishing wallets were found to share 61 intermediaries.

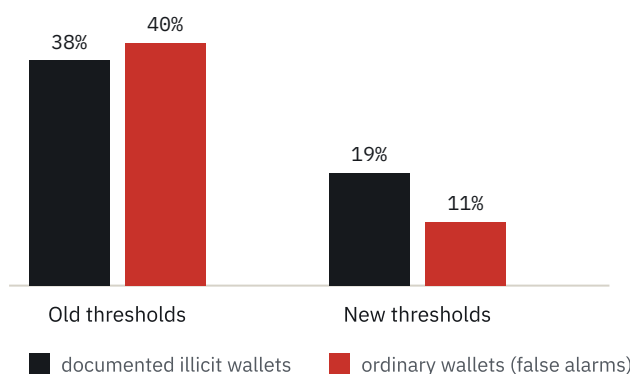
Evidence hashing

Every provider response hashed on arrival; the report sealed with a content hash.

A case file UI

The results view answers the officer's questions in order; a dark console skin by default.

THE AMOUNT-SPLIT RULE, MEASURED ON 79 REAL WALLETS



THE INCONVENIENT RESULT, PUBLISHED

The old thresholds flagged 40% of ordinary wallets. The new ones flag 11% — and no threshold separates the two groups.

So a pattern is a reason to look closer and is never scored. 40 documented-illicit and 48 ordinary wallets were fetched by script, 1,416 provider requests snapshotted so the run reproduces offline, 68 threshold combinations swept.

16–20 SEPTEMBER

Phases 4–6 · becoming Exchequer, and showing it

PHASE 4 · 16 SEP · 23 COMMITS

TraceChain → Exchequer

Renamed at 00:15 because a rival SIH26183 team ships as “TraceChain”. The Exchequer was named for the chequered cloth on which the Crown’s money was counted — one square red, the sum being traced.

Hosted on Railway

One Docker image, a /data volume. Sign-in added, then removed at the user’s request.

Instant demos

Responses kept on disk and the demos traced at startup: flagship 43 s cold → 0.12 s.

CI on every push

Tests and the frontend build; a live pick from Etherscan’s phishing list; “Stop after 1:30”.

PHASE 5 · 17 SEP · 7 COMMITS



The chequer mark. A sign-in page drawn over the real ledger of addresses the tool knows, coloured by what it knows. **Present mode:** any case as four projector screens. Tron and BSC pagination bounded by records read, after a busy wallet paged for minutes on the hosted demo. 234 tests.

PHASE 6 · 18–20 SEP · DOCUMENTS AND THE LAUNCH FILM

19 September: the **Director proposal** (5 pages) and the **Research Report** (12 pages, with RESEARCH.md as an 18-page appendix). 19–20 September: the launch film in three cuts — v1 screenshots (rejected as bland), v2 “one red square”, and **v3, 43.2 seconds, the latest:** drawn SVG and type only, narrated by the Emma voice, with its own score.

V3 STORYBOARD · 43.2 S

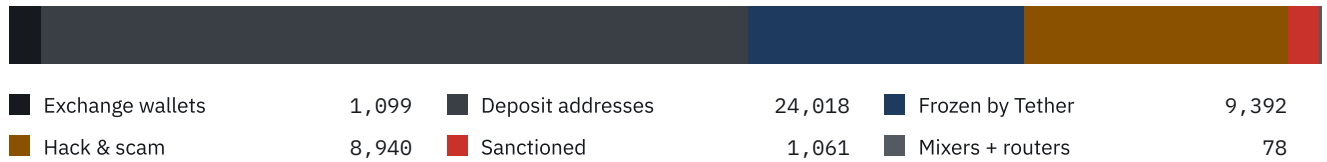
The report	The trace	The map	Screened	3 complaints	The handoff	The stance	Exchequer
5.3 s	7.4 s	4.7 s	4.2 s	5.3 s	6.8 s	5.3 s	4.2 s

“No model, no clustering, no proprietary score. But everything hand-checked. That’s Exchequer for you in a nutshell.”

22-23 SEPTEMBER · 11 COMMITS · 4 DEPLOYMENTS

Phase 7 · from 407 labels to 44,588

WHAT THE 44,588 LABELLED ADDRESSES ARE



23 SEPTEMBER, COMMIT BY COMMIT

00:05	2dda364	Exchange labels 337 → 1,020 · OFAC refreshed	
00:12	3053ab5	Stolen-funds category · mixer pools rebuilt	
00:24	f825542	Five more governments' sanctions lists	DEPLOYED
00:40	966ffec	All demos re-verified cold	DEPLOYED
03:21	53740c1	8,600 reported phishing and scam wallets	
05:25	8803efb	19,027 Bitget customer deposit addresses	DEPLOYED
06:55	1971e33	README rewritten · every data source documented	
07:06	318426a	Tether's USDT freeze list · demo 11	
07:13	2dcdec1	Swaps into USDT/USDC followed onward	
08:05	073b1c6	Draft request letters to exchange and Tether	
08:09	68ba2af	4,991 Binance customer deposit addresses	DEPLOYED

CAUGHT BEFORE IT SHIPPED

EigenLayer operator tags read as exchanges · a verified Tron wallet dropped by a re-scan · an exchange count first given as 106 (really 92) · a Tornado pool Tether froze losing its mixer label.

HOW EVERY ADDRESS GOT IN

Each import names a pinned source, checks every address on chain before accepting it, and records where it came from. Government lists are copied exactly as published.

23 SEPTEMBER · MORNING

Phase 8 · the bug hunt, and where it stood

1. Static

ruff over backend, scripts and tests



2. Endpoints

every endpoint on 15 real cases, error paths too



3. Browser

every page in headless Brave; errors captured

BUG 1 A false “Frozen by Tether”

Tether’s blacklist holds the zero address and 24 system addresses where tokens are burned. A USDC burn was flagged as a freeze and offered a letter about the zero address. Fixed in the importer and the matcher.

BUG 2 “Stop after 1:30” did not hold

Each swap follow-on got a fresh time budget, so a case could run 4.5 minutes. Follow-ons now share what is left.

Zero JavaScript errors across 15 cases. Committed as 0b46c4e at 08:26 and deployed the same morning. Two database backups swept into that commit by mistake were taken out before anything left the machine.

WHAT EXISTED THAT MORNING

Tracing 3 chains · 8 assets · both directions · the time rule · swaps into stablecoins followed

Attribution 25,117 exchange addresses · 24,018 customer deposit addresses · probable deposits inferred

Screening 6 governments · Tether freezes · mixers stop the trace · hack and scam wallets

Outputs case view · Present mode · sealed report · draft letters to the exchange and Tether

ALSO

RESEARCH.md sources every claim and every dataset · DEMO.md: 11 verified traces · JUDGE_QA.md: 15 answers · 273 tests · CI on every push · the one-pager, two PDFs and the launch film.

OPEN ITEMS THAT MORNING

- ☒ Push and deploy the bug fix 0b46c4e
- ☐ Licences before commercial use: OpenSanctions (non-commercial), ScamSniffer (GPL-3.0)
- ☐ Demo order: skip trace 2; trace 11 answers “why Tron?”
- ☐ Letters: an officer confirms the legal provision
- ☐ SIH registration as Exchequer; the deck still says TraceChain
- ☐ Still to build: live TronScan tags on Tron traces; Bitcoin; a PDF report

23 SEPTEMBER · AFTERNOON AND NIGHT · 11 COMMITS · 5 DEPLOYMENTS

Phase 9 · wider, and honest

COMMIT BY COMMIT

11:46	87e02bd	TronScan's own tag read live for Tron wallets no file names	
12:03	f9bd400	Faster tag lookups · mistyped Tron addresses caught by checksum	DEPLOYED
12:08	cd4e113	Repository tidied · proposal PDFs and the launch film committed	
12:12	f9ded52	Judge Q&A to 18 answers · demo 12, an exchange found live	
12:19	5af080a	TronScan warning tags ("Suspicious") become screening hits	
12:21	2a5cba2	Related cases search follow-ons and flagged-but-unnamed wallets	
12:56	a4a842a	Polygon and Arbitrum — five chains	DEPLOYED
13:28	8df02c2	BSC: quiet wallets found by nonce · 38 → 184 labels · wallet statement	DEPLOYED
21:02	8558ab6	Pro-rata tracing: how much of the money is the victim's	DEPLOYED
21:16	b5eed01	Demos re-run every morning by CI · the amount in the exchange letter	DEPLOYED
21:23	3e3fe05	Demo expectations recorded: 12 of 12 as documented	

PRO-RATA, ON A REAL BSC WALLET



WHAT EXISTS NOW

Tracing	5 chains · native coins, USDT, USDC · both directions · the time rule · swaps followed
Attribution	25,508 exchange addresses · TronScan tags read live · deposits inferred · a pro-rata amount
Screening	6 governments · Tether freezes · mixers stop the trace · scam wallets · TronScan warnings
Outputs	case view · wallet statement · Present mode · sealed report · letters that name the amount

DEMO.md: 12 traces, checked every morning · JUDGE_QA.md: 19 answers · 311 tests · the site stays up for judges.

OPEN ITEMS

- ☐ Licences before commercial use: OpenSanctions (non-commercial), ScamSniffer (GPL-3.0)
- ☐ Letters: an officer confirms the legal provision
- ☐ Old BSC wallets take about a minute on the free provider
- ☐ Still to build: Bitcoin · a PDF report · pro-rata on reverse traces
- ☐ The 25-second social cut of the film · Railway config before 1 Dec